SWISS-MODEL Homology Modelling Report

Model Building Report

This document lists the results for the homology modelling project "Trastuzumab_Fv_HER2_Model" submitted to SWISS-MODEL workspace on March 23, 2026, 6:48 a.m..The submitted primary amino acid sequence is given in Table T1.

If you use any results in your research, please cite the relevant publications:

- Waterhouse A, Bertoni M, Bienert S, Studer G, Tauriello G, Gumienny R, Heer FT, de Beer TAP, Rempfer C, Bordoli L, Lepore R, Schwede T
  SWISS-MODEL: homology modelling of protein structures and complexes.
  Nucleic Acids Res 46, W296-W303. (2018) 29788355 10.1093/nar/gky427
- Bienert S, Waterhouse A, de Beer TAP, Tauriello G, Studer G, Bordoli L, Schwede T
  The SWISS-MODEL Repository - new features and functionality.
  Nucleic Acids Res 45, D313-D319. (2017) 27899672 10.1093/nar/gkw1132
- Studer G, Tauriello G, Bienert S, Biasini M, Johner N, Schwede T
  ProMod3 - A versatile homology modelling toolbox.
  PLOS Comp Biol 17(1), e1008667. (2021) 33507980 10.1371/journal.pcbi.1008667
- Studer G, Rempfer C, Waterhouse AM, Gumienny R, Haas J, Schwede T
  QMEANDisCo - distance constraints applied on model quality estimation.
  Bioinformatics 36, 1765-1771. (2020) 31697312 10.1093/bioinformatics/btz828
- Bertoni M, Kiefer F, Biasini M, Bordoli L, Schwede T
  Modeling protein quaternary structure of homo- and hetero-oligomers beyond binary interactions by homology.
  Scientific Reports 7. (2017) 28874689 10.1038/s41598-017-09654-8

Results

The SWISS-MODEL template library (SMTL version 2026-03-18, PDB release 2026-03-13) was searched with BLAST (Camacho et al.) and HHblits (Steinegger et al.) for evolutionary related structures matching the target sequences in Table T1. For details on the template search, see Materials and Methods. Overall 50 templates were found (Table T2).

Models

The following model was built (see Materials and Methods "Model Building"):

Model #01	File	Built with	Oligo-State	Ligands	GMQE	QMEANDisCo Global
	PDB	ProMod3 3.6.0	hetero-1-1-mer	None	0.86	0.82 ± 0.05

Template	Seq Identity	Oligo-state	QSQE	Found by	Method	Resolution	Seq Similarity	Range	Coverage	Description
6dez.1	89.63	hetero-1-1-mer	0.86	BLAST	X-ray	3.20Å	0.57	A: 1-212 B: 1-220	1.00	Anti-phosphotyrosine antibody PY20-4D5 light chain Anti-phosphotyrosine antibody PY20-4D5 heavy chain

Excluded ligands

Ligand Name.Number	Reason for Exclusion	Description
SO4.1	Not biologically relevant.	SULFATE ION

```
Target     DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSASFLYSGVPSRFSGSRSGTDFTLTISSLQP
6dez.1.A   DIQMTQSPSSLSASVGDRVTITCSASQGISNYLNWYQQKPGKAPKLLIYYTSSLHSGVPSRFSGSRSGTDFTLTISSLQP

Target     EDFATYYCQQHYTTPPTFGQGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQ
6dez.1.A   EDFATYYCQQYSKVPWTFGQGTKVEIKRTVAAPSVFIFPPSDSQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQ

Target     ESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
6dez.1.A   ESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
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Target    EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARIYPTNGYTRYADSVKGRFTISADTSKNTAY
6dez.1.B  EVQLVESGGGLVQPGGSLRLSCAASGYTFTEYTMHWVRQAPGKGLEWMGGINPNSGGTRDNQRFKGRFTISADTSKNTAY

Target    LQMNSLRAEDTAVYYCSRWGGDG----FYAMDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEP
6dez.1.B  LQMNSLRAEDTAVYYCARRGPYGNYYNSYYFDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEP

Target    VTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEP
6dez.1.B  VTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEP
```

Materials and Methods

Template Selection

For each identified template, the template's quality has been predicted from features of the target-template alignment. The templates with the highest quality have then been selected for model building.

Model Building

Models are built based on the target-template alignment using ProMod3 (Studer et al.). Coordinates which are conserved between the target and the template are copied from the template to the model. Insertions and deletions are remodelled using a fragment library. Side chains are then rebuilt. Finally, the geometry of the resulting model is regularized by using a force field.

Model Quality Estimation

The global and per-residue model quality has been assessed using the QMEAN scoring function (Studer et al.).

Ligand Modelling

Ligands present in the template structure are transferred by homology to the model when the following criteria are met: (a) The ligands are annotated as biologically relevant in the template library, (b) the ligand is in contact with the model, (c) the ligand is not clashing with the protein, (d) the residues in contact with the ligand are conserved between the target and the template. If any of these four criteria is not satisfied, a certain ligand will not be included in the model. The model summary includes information on why and which ligand has not been included.

Oligomeric State Conservation

The quaternary structure annotation of the template is used to model the target sequence in its oligomeric form. The method (Bertoni et al.) is based on a supervised machine learning algorithm, Support Vector Machines (SVM), which combines interface conservation, structural clustering, and other template features to provide a quaternary structure quality estimate (QSQE). The QSQE score is a number between 0 and 1, reflecting the expected accuracy of the interchain contacts for a model built based a given alignment and template. Higher numbers indicate higher reliability. This complements the GMQE score which estimates the accuracy of the tertiary structure of the resulting model.

References

- Camacho C, Coulouris G, Avagyan V, Ma N, Papadopoulos J, Bealer K, Madden TL
  BLAST+: architecture and applications.
  BMC Bioinformatics, 10, 421-430. (2009) 20003500 10.1186/1471-2105-10-421
- Steinegger M, Meier M, Mirdita M, Vöhringer H, Haunsberger SJ, Söding J
  HH-suite3 for fast remote homology detection and deep protein annotation.
  BMC Bioinformatics 20, 473. (2019) 31521110 10.1186/s12859-019-3019-7

Table T1:

Primary amino acid sequences for which templates were searched and models were built.

```
DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSASFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQ
GTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQG
LSSPVTKSFNRGEC

EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARIYPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWG
GDGFYAMDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQT
YICNVNHKPSNTKVDKKVEP
```

Table T2:

Template	Seq Identity	Oligo-state	QSQE	Found by	Method	Resolution	Seq Similarity	Coverage	Description
6dez.1	89.63	hetero-1-1-mer	0.86	BLAST	X-ray	3.20Å	0.57	1.00	Anti-phosphotyrosine antibody PY20-4D5 light chain; Anti-phosphotyrosine antibody PY20-4D5 heavy chain
8t7f.4	92.63	hetero-1-1-mer	0.83	BLAST / HHblits	X-ray	3.50Å	0.58	1.00	S1 variant of Fab F1 light chain; S1 variant of Fab F1 heavy chain
8t7f.2	92.63	hetero-1-1-mer	0.83	BLAST / HHblits	X-ray	3.50Å	0.58	1.00	S1 variant of Fab F1 light chain; S1 variant of Fab F1 heavy chain
6dez.2	89.63	hetero-1-1-mer	0.83	BLAST	X-ray	3.20Å	0.57	1.00	Anti-phosphotyrosine antibody PY20-4D5 light chain; Anti-phosphotyrosine antibody PY20-4D5 heavy chain
1n8z.1	100.00	hetero-1-1-mer	0.90	HHblits	X-ray	2.52Å	0.62	1.00	Herceptin Fab (antibody) - light chain; Herceptin Fab (antibody) - heavy chain
7kfy.1	89.61	hetero-1-1-mer	0.90	BLAST / HHblits	X-ray	2.16Å	0.58	1.00	light chain of human antibody C1A-F10 Fab; heavy chain of human antibody C1A-F10 Fab
4hs8.1	89.35	hetero-1-1-mer	0.89	BLAST / HHblits	X-ray	2.60Å	0.57	1.00	antibody hu5B3.v2 Fab light chain; antibody hu5B3.v2 Fab heavy chain
6and.1	89.17	hetero-1-1-mer	0.88	BLAST / HHblits	X-ray	1.75Å	0.57	1.00	Pinatuzumab Fab light chain; Pinatuzumab Fab Heavy chain
3kr3.1	89.17	hetero-1-1-mer	0.88	BLAST / HHblits	X-ray	2.20Å	0.57	1.00	antibody-Fab (light chain); antibody-Fab (heavy chain)
1tzi.1	93.55	hetero-1-1-mer	0.89	BLAST / HHblits	X-ray	2.80Å	0.58	1.00	Fab YADS2 Light Chain; Fab YADS2 Heavy Chain
6uta.1	88.94	hetero-1-1-mer	0.89	BLAST	X-ray	3.10Å	0.57	1.00	Z004 iGL Fab light chain; Z004 iGL Fab heavy chain
8yns.1	92.86	hetero-1-1-mer	0.90	BLAST	EM	NA	0.59	1.00	Antibody 1B3 Fab light chain; Antibody 1B3 Fab heavy chain
3skj.1	88.25	hetero-1-1-mer	0.87	BLAST / HHblits	X-ray	2.50Å	0.57	1.00	Antibody, light chain; Antibody, heavy chain
7chy.1	93.09	hetero-1-1-mer	0.87	BLAST / HHblits	X-ray	2.65Å	0.59	1.00	light chain of antibody binding fragment of IgG26; heavy chain of antibody binding fragment of IgG26
4zfg.1	93.32	hetero-1-1-mer	0.87	BLAST	X-ray	2.27Å	0.58	1.00	Fragment antigen binding 5A12 light chain; Fragment antigen binding 5A12 heavy chain
4k7p.1	88.91	hetero-1-1-mer	0.87	BLAST / HHblits	X-ray	2.95Å	0.57	1.00	antibody rhumAb6 Fab fragment light chain; antibody rhumAb6 Fab fragment heavy chain
7s0b.1	90.30	hetero-1-1-mer	0.86	BLAST / HHblits	X-ray	2.90Å	0.58	1.00	N-612-056 Light Chain; N-612-056 Fab Heavy Chain
1tzh.1	93.07	hetero-1-1-mer	0.86	BLAST	X-ray	2.60Å	0.59	1.00	Fab YADS1 Light Chain; Fab YADS1 Heavy Chain
9iya.1	93.32	hetero-1-1-mer	0.87	BLAST / HHblits	EM	NA	0.59	1.00	anti-BRIL Fab Light chain; anti-BRIL Fab Heavy chain
1tzh.1	93.29	hetero-1-1-mer	0.85	BLAST / HHblits	X-ray	2.60Å	0.59	1.00	Fab YADS1 Light Chain; Fab YADS1 Heavy Chain
1tzh.1	92.84	hetero-1-1-mer	0.85	BLAST / HHblits	X-ray	2.60Å	0.58	1.00	Fab YADS1 Light Chain; Fab YADS1 Heavy Chain

Template	Seq Identity	Oligo-state	QSQE	Found by	Method	Resolution	Seq Similarity	Coverage	Description
1tzh.1	93.06	hetero-1-1-mer	0.84	HHblits	X-ray	2.60Å	0.58	1.00	Fab YADS1 Light Chain; Fab YADS1 Heavy Chain
9j05.1	92.86	hetero-1-1-mer	0.85	BLAST	EM	NA	0.59	1.00	Antibody 1b3 Fab Light chain; Antibody 1b3 Fab Heavy chain
5xhv.1	94.20	hetero-1-1-mer	0.83	BLAST / HHblits	X-ray	3.35Å	0.59	0.99	S40 light chain; S40 heavy chain
5xhv.1	92.82	hetero-1-1-mer	0.82	HHblits	X-ray	3.35Å	0.58	1.00	S40 light chain; S40 heavy chain
7s0c.1	88.94	hetero-1-1-mer	0.82	BLAST / HHblits	EM	NA	0.57	1.00	N-612-017 Light Chain; N-612-017 Fab Heavy Chain
2fjh.1	93.09	hetero-2-2-mer	0.51	BLAST / HHblits	X-ray	3.10Å	0.59	1.00	Fab fragment light chain; Fab fragment heavy chain
7mk6.1	91.20	hetero-4-4-mer	0.48	BLAST / HHblits	X-ray	3.10Å	0.58	1.00	Fab light chain; Fab heavy chain
8hpv.1	89.35	hetero-3-3-mer	0.41	BLAST / HHblits	EM	NA	0.57	1.00	fab L5.34; fab L5.34
7s0d.1	89.84	hetero-1-1-mer	0.80	BLAST / HHblits	EM	NA	0.57	1.00	N-612-014 Light Chain; N-612-014 Fab Heavy Chain
7czr.1	90.68	hetero-2-2-mer	0.45	BLAST / HHblits	EM	NA	0.58	0.99	IGL c4203_light_IGKV1-9_IGKJ4,IGL c4203_light_IGKV1-9_IGKJ4,Uncharacterized protein; IG c542_heavy_IGHV3-53_IGHD3-10_IGHJ6,IGH@ protein
8vr9.1	92.58	hetero-1-1-mer	0.66	BLAST	EM	NA	0.58	0.99	R023 Fab light chain; R023 Fab heavy chain

The table above shows the top 32 filtered templates. A further 17 templates were found which were considered to be less suitable for modelling than the filtered list.
1tzi.1, 2fjh.1, 3skj.2, 4zff.1, 5xhv.2, 6and.1, 6dez.1, 6dez.2, 6oge.1, 6uta.2, 7chy.1, 7mk6.1, 8q6j.1, 8t7f.4, 8vrb.1, 8yns.1, 9j05.1